

ANIMAGUS: A Provably Secure Accordion Mode of Operation

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Abstract

Block ciphers are designed to operate on fixed length blocks of bits. A block cipher mode of operation is used in order to encrypt variable-length input. These modes process multiple data blocks and ensure information security through the application of block cipher algorithms. While there are several NIST-approved block cipher modes, they exhibit certain inherent limitations and security vulnerabilities. Hence, the necessity for a novel mode has emerged. NIST aims to design an accordion mode, characterized as a tweakable variable-input-length strong pseudorandom permutation, offering enhanced security and performance attributes compared to existing modes. It lastly proposed to develop accordion mode types from different variants of HCTR2 algorithm. Thus, we propose a new design based on HCTR2 structure for an accordion mode in this article. The proposed design accommodates variable-length inputs while ensuring strong pseudorandom permutation security. It also exhibits adaptability with any underlying block cipher and facilitates variable-length tweaks. This ultimately provides a more versatile framework compared to numerous existing schemes in the literature.

Keywords: Block ciphers, accordion mode, tweakable, variable input length.

1 Introduction

Block cipher modes of operation represent a suite of techniques employed to extend the applicability of block ciphers to handle variable-length input. These modes of operation by processing multiple contiguous blocks of data, thereby providing information security predicated on the underlying block cipher. Modes of operation, often abbreviated as modes, also facilitate functionalities such as authentication, storage encryption, and the adaptation of block ciphers into stream ciphers with different modes. There are several NIST-approved block cipher modes which are explained in [11–17]. Over time, these established modes have been identified as exhibiting limitations, and in certain instances, have been deemed to possess inadequate security properties. Consequently, NIST initiated a thorough review of these modes [24] and planned a workshop.

The Third Workshop on Block Cipher Modes of Operation organized by NIST in October 2023 served as a platform for the critical evaluation of the limitations, security characteristics, and efficiency of the currently approved modes. As a result of this workshop, NIST Internal Report (IR) 8459 [24] was published in 2024. This report provides an analysis of each mode separately, with a significant emphasis on the potential vulnerabilities arising from the treatment of the initialization vector, plaintext length, and block size. The report concludes with a set of recommendations, including the proscription of the Electronic Codebook (ECB) mode [11], the proposal of specific revisions to modes such as Counter with CBC-MAC (CCM) [13] and Galois/Counter Mode (GCM) [15] to enhance their security posture, and the standardization of a new mode to prevent potential misuse scenarios. As a direct consequence of these findings, NIST looks for a novel mode designated as the Accordion Cipher Mode, also known as the accordion mode.

The requirements for the accordion mode were formally articulated in a proposal draft published in April 2024 [5]. This document outlined the general construction and anticipated properties of the accordion mode as a preliminary framework for requesting feedback from the cryptographic community. According to this initial proposal, the accordion mode should be a tweakable variable-input-length strong pseudorandom permutation exhibiting security and performance advantages over existing modes.

These advantages encompass features such as nonce misuse resistance, nonce hiding, release of unverified plaintext, and beyond birthday bound security, among others.

Following the dissemination of this draft detailing the general structure and requirements, NIST hosted a workshop in June 2024, the NIST Work-

shop on the Requirements for an Accordion Cipher Mode, to facilitate in-depth discussion of these properties and to gather comprehensive feedback. The report summarizing the proceedings of this workshop was subsequently published in November 2024 [27]. During this workshop, the initial requirements underwent refinement based on the collective needs expressed by participants from diverse sectors, aiming to achieve a consensus on the final specifications. The culmination of these extensive studies resulted in the issuance of Requirements for Cryptographic Accordions [6] in April 2025. This document formally announced the final requirements for the accordion mode, specifying its architectural framework as a three-layer construction comprising derived functions, the accordion mode itself, and the underlying block cipher, with these requirements classified and illustrated systematically according to these layers.

As indicated in the previously mentioned accordion mode reports and presentations delivered at the dedicated workshop, while there exist certain cryptographic algorithms documented in the literature like the intended accordion mode, they need adaptation and further development.

In this article, we present our proposed design for this new mode of operation and provide a formal security proof to substantiate its security claims. The proposed design accommodates variable-length inputs, with ensuring strong pseudorandom permutation (SPRP) security, eliminating the requirement for padding within its specified minimum and maximum message length constraints. This characteristic differentiates it from most block cipher modes, which typically necessitate padding to align message sizes. Furthermore, our construction exhibits adaptability with any underlying block cipher, consequently supporting a wide range of block and key sizes. Our design allows application across all required areas with encode-then-encipher method, catering to diverse user needs. The design also facilitates variable-length tweaks by preprocessing the tweak data prior to its integration into the Accordion Mode. This preprocessing mechanism, concurrently provides a significantly expanded range for the tweak size. Ultimately, our design offers a more versatile framework compared to numerous existing schemes in the literature, critically achieving this enhanced flexibility without compromising security guarantees. In the paper, Section 2 is dedicated to an overview of the existing research. We detail our design in Section 3, providing its formal specification, outlining its parameters, and identifying its relevant applications. Section 4 presents the security analysis of our design, including its formal proof of security.

2 Previous Works

Within NIST reports, illustrative design methodologies are presented concerning the primary requirement of tweakable variable-input-length strong pseudorandom permutations. These methodologies invariably incorporate the utilization of a tweak within their algorithmic structures and employ two principal construction paradigms to accommodate variable-length input: the hash-encrypt-hash and the encrypt-mix-encrypt constructions. Nevertheless, a subset of these approaches lacks universal applicability across arbitrary block ciphers, and furthermore, they do not uniformly guarantee the satisfaction of all stipulated security properties. In this section, we present some examples of these approaches.

In 2003, Halevi and Rogaway introduced a tweakable enciphering scheme named CMC [19], which pioneered the encrypt-mix-encrypt methodology. This approach employs a mixing function centered between two encryption processes to ensure inter-block dependency. Specifically, Halevi and Rogaway utilized the Cipher Block Chaining (CBC) mode of operation within both encryption layers, while achieving inter-block mixing by incorporating a value derived from other blocks into the plaintext blocks. Notably, the encryption and decryption procedures in CMC are analogous, with the sole distinction residing in the application of the inverse of the encryption algorithm during decryption. This inherent symmetry is a beneficial attribute for establishing strong pseudorandom permutation security, a property demonstrated by their algorithm. Halevi and Rogaway further advanced this methodology with the design of the EME algorithm, which also based on the encrypt-mix-encrypt method but offers distinct advantages over CMC. EME utilizes the Electronic Codebook (ECB) mode for its encryption layers. Compared to CMC, it offers advantages such as parallelizability, single-key operation, and variable input length.

Another design predicated on the encrypt-mix-encrypt paradigm is an authenticated encryption scheme, AEZ, submitted by Krovetz and Rogaway to the CAESAR Competition in 2017 [20]. Although conceived for an authenticated encryption competition, AEZ functions as an arbitrary-input-length cipher incorporating a tweak. Consequently, it has been proposed as a potential design strategy for the accordion mode. AEZ leverages a Feistel construction within its encryption layers; however, a less desirable characteristic for accordion mode design is its utilization of a reduced-round Advanced Encryption Standard (AES) as the round function as noted in a workshop presentation.

The accordion mode proposal also cites ZCZ [2], a strong pseudorandom

permutation proposed by Bhaumik, List, and Nandi in 2018, as another illustrative example of the encrypt-mix-encrypt method with tweakable block cipher, exhibiting substantial structural similarities to AEZ. Notably, ZCZ achieves security beyond the birthday bound, a characteristic not commonly found in the majority of algorithms in the literature, alongside indistinguishability from a pseudorandom permutation.

Building upon the Luby and Rackoff [22] structure, Naor and Reingold proposed an innovative approach in [25] to extend the applicability of block ciphers from fixed-length to variable-length input, giving rise to the hash-encrypt-hash methodology. The fundamental concept involves partitioning the plaintext into two segments: one of a fixed block size and the remaining portion. Subsequently, the larger segment is hashed and the resultant digest is combined with the smaller segment. The smaller segment is then encrypted, and a keystream is generated to be mixed with the larger segment.

HCTR, proposed by Wang, Feng, and Wu in 2005 [28], serves as an illustrative example of a scheme employing the hash-encrypt-hash method. It transforms a conventional block cipher into a tweakable variable-input-length cipher. Provided that the underlying block cipher utilized with HCTR exhibits strong pseudorandom permutation properties, HCTR can achieve indistinguishability from a strong pseudorandom permutation. The core structure of HCTR has been a foundation for other tweakable variable-input-length ciphers, such as PEP [4], HCH [3], and TET [18], which modify the original design with different encryption methods, hashes, or other components. These schemes typically rely on traditional block cipher modes of operation like ECB, CTR, and OFB for keystream generation. Adiantum, a tweakable length-preserving encryption algorithm by Crowley and Biggers (2018), differs from this traditional approach. Instead of using a block cipher mode for keystream generation, Adiantum utilizes a stream cipher. It employs a hash-block cipher-stream cipher-hash (HBSH) construction, which makes it a unique example within the realm of variable-input-length ciphers.

In 2013, Shrimpton and Terashima introduced TCT_2 [26] as a variable-input-length tweakable cipher within their Protected-IV (PIV) framework. This scheme uses a three-round unbalanced Feistel structure to combine fixed-input-length tweakable block ciphers with variable-length components. TCT_2 is based on a hash-encrypt-hash style tweakable block cipher, which is iterated to improve security. It also incorporates counter-mode encryption and a hash-based tweak extension to achieve beyond birthday-bound (BBB) security. In 2019, Dutta and Nandi proposed Tweakable HCTR [9], a modification of the original HCTR. It achieves BBB security by replacing all encryption algorithms in HCTR with tweakable encryption algorithms.

However, a drawback of Tweakable HCTR is that its security degrades when tweaks are reused. In 2024, Datta et al. introduced tweakable block cipher based accordion mode HCTR+ [8], which also provides BBB security even with repeated tweaks. This is a significant improvement over Tweakable HCTR. A key challenge highlighted by these examples is that while they demonstrate how to achieve BBB security using tweakable block ciphers, the NIST is seeking a solution that provides the same level of security using the widely adopted AES which is not a tweakable block cipher.

The ACCOR algorithms [10], presented by Duy et al. at the NIST Accordion Mode Workshop, are also based on the HCTR scheme and follow the hash-encrypt-hash structure. This study introduced two distinct algorithms: ACCOR-S and ACCOR-L. ACCOR-S algorithm is optimized for high performance and provides birthday-bound security. ACCOR-L algorithm is designed to achieve beyond birthday-bound security, though with lower performance. Both ACCOR algorithms are built upon AES and require a minimum of three keys, all of which are derived from a single master key.

In 2021, Crowley, Huckleberry, and Biggers introduced HCTR2 [7] as a more robust version of the original HCTR scheme. The primary difference between the two is the hash function. The authors of HCTR2 claimed that the hash function used in the original HCTR was not almost-XOR-universal, and they addressed this vulnerability in their new design. Beyond fixing this security flaw, HCTR2 also brought several other improvements: enhanced security bound, variable-length tweak and reduced key count. These modifications position HCTR2 as a superior alternative to its predecessor. According to the announcement published in June 2025, NIST proposed that different variants be developed from HCTR2 for the desired accordion mode types.

Our design employs the hash-encrypt-hash structure, following the HCTR2 model as recommended by NIST, due to its balance of performance and flexibility. A key departure from other schemes that we have discussed is our use of a cryptographic hash algorithm instead of a universal hash. While this may incur a performance overhead, it enhances security by providing collision resistance. Specifically, we use the HMAC algorithm, which is widely used in the industry, though alternatives like KMAC, as suggested for SHA3, could also be employed. A second distinguishing feature of our design is the inclusion of a secondary encryption branch. This modification facilitates the implementation of authenticated encryption by allowing the first data block to be designated as fixed authentication data. This approach maintains the same level of security and prevents forged message authentication without

altering the core encryption method.

We opted for a birthday-bound security level for our design. While beyond-birthday-bound security is a requirement for accordion mode application utilizing AES, its high performance overhead and implementation complexity render it less practical for many use cases. Simpler modes of operation, which facilitate more straightforward security analysis, are often favored, especially for large block size encryption. Therefore, our algorithm was designed with birthday-bound security to be a flexible solution adaptable to various block ciphers and input sizes. We recognize the potential for future work, and a version providing beyond-birthday-bound security is developed in subsequent studies.

Our design, named **Animagus**, is summarized in Table 1 alongside other related algorithms. The table, taken from [10], has been updated to include a range of more similar schemes, including newly released ones. It is divided into two sections: the first features algorithms based on the encrypt-mix-encrypt method, and the second lists those using the hash-encrypt-hash method. Our design, located in the final row, demonstrates enhanced security with a relatively minor increase in computational operations.

3 Our Accordion Design

We propose a new accordion mode called **Animagus** which is designed using the hash-encrypt-hash construction paradigm. Animagus is built upon a block cipher $E : \{0, 1\}^n \times K \rightarrow \{0, 1\}^n$ characterized by an n -bit block size and a k -bit key size, and the HMAC keyed hash function employing SHA3 to produce a $2n$ -bit output. The message space is defined as $\mathcal{M} = \{0, 1\}^{na}$ where a represents the maximum number of n -bit blocks in the plaintext. Although the specified minimum plaintext length is $2n$ -bit, the algorithm accommodates messages of any length greater than or equal to this value. A key design feature is that the message length is not required to be a multiple of n . Furthermore, the design’s operational mechanism remains consistent for messages of exactly $2n$ bits. This is because all inputs are invariably processed alongside supplementary data, such as tweaks or authentication information, ensuring the algorithm’s full functionality is always engaged. The key space of Animagus is inherited from the underlying block cipher E . The key provided to the algorithm is utilized for both the block cipher E and the HMAC function. A schematic representation of the Animagus encryption process is depicted in Figure 1, where E denotes the encryption algorithm, H represents HMAC, $P = P_1 || P_2 || \overline{P}$ signifies the plaintext, and $C = C_1 || C_2 || \overline{C}$

Table 1: Summary of Previous Works

Design	Security	Cipher	Hash	Operation Count
CMC	$7q^2a^2/2^n$	BC		$(2m+1)BC$
EME	$7q^2a^2/2^n$	BC		$(2m+2)BC$
AEZ	$13q^2a^2/2^{128}$	BC		mBC
ZCZ	$3q^2a^2/2^{2n+1}$	TBC	Polynomial	$(2m+6+m/n)TBC + 4HF$
HCTR	$4.5q^2a^2/2^n$	BC	Polynomial	$mBC + 2HF$
PEP	$3q^2a^2/2^n$	BC	Polynomial	$(m+5)BC + (4m-6)FM$
HCH	$7q^2a^2/2^n$	BC	Polynomial	$(m+3)BC + 2(m-1)FM + 2HF$
TET	$1.5q^2a^2/2^n$	BC	Invertible Block-wise Universal	$(m+1)BC + 2mFM$
TCT ₂	$qa^2/2^n + 6q^3a^3/(2^{2n-2} - q^3a^3) + 6^4q^3/(2^{2n-2} - 6^3q^3)$	BC	Polynomial, AU/AXU	$(6m+6)BC + (7m+7)HF$
THCTR	$2qa/2^n + 2q^2a/2^{n+s}$	TBC	n -bit keyed AXU, $(n+s)$ -bit keyed pAXU	$mTBC + 3HF$
ADIANTUM	$q^2a/2^{116}$	BC	Polynomial, ϵ - Δ U	$BC + SC + FM + FA + 2HF$
HCTR2	$1.5q^2a^2/2^n$	BC	Polynomial, AXU	$mBC + 2HF$
HCTR+	$183q^2/(2^n - 1)^2 + 2\min\{4q^2a, q^2a^2\}/2^{2n} + 3q^2/2^{2n} + 10/2^n$	TBC	$2n$ -bit keyed AXU	$(m+2)TBC + 2HF$
ACCOR-S	$4.5q^2a^2/2^{128}$	BC	Polynomial, AXU	$mBC + 2HF$
ACCOR-L	$qa/2^{124}$	BC	Polynomial, AXU	$(m+1)BC + 4FM + 2HF$
ANIMAGUS	$q^2a^2/2^n$	BC	Cryptographic	$(m+2)BC + 2HF$

BC: Block Cipher, TBC: Tweakable Block Cipher, SC: Stream Cipher, HF: Hash Function, AU: Almost Universal, AXU: Almost XOR Universal, pAXU: partial AXU, Δ U: Almost Δ Universal, FM: Field Multiplication, FA: Field Addition, n : block size, a : maximum message length in blocks, q : number of queries, s : tweak length after hashing, m : number of blocks of a specific message

denotes the corresponding ciphertext. The computation of the tweak T is performed within a derived function, the specifics of which are contingent upon the application type of the accordion mode and will be elaborated upon following the detailed explanation of the Animagus construction. In this section, we provide a comprehensive explanation of our design, outlining its operational principles, defining its parameters, and identifying its application

areas.

The Animagus algorithm processes a plaintext P of variable length, adhering to the message size specifications previously defined. This plaintext is segmented into three components: the initial two branches, P_1 and P_2 , which are of equal length, corresponding to the n -bit block size of the underlying encryption algorithm, and the remaining segment, denoted as $\overline{P}$. The algorithm subsequently processes T and $\overline{P}$ through an HMAC function, utilizing the concatenation $\overline{P} \parallel 10 \parallel T$ to ensure robust domain separation. If the message length is $2n$ bits, the HMAC only hashes $10 \parallel T$ as a third branch. The resulting $2n$ -bit hash output is split into two n -bit parts, A_1 and A_2 . These parts are XORed with the initial plaintext branches to ensure the intermixing of this third plaintext segment with the first two. The initial n bits of the resulting $2n$ -bit hash value, A_1 , are added to the second branch P_2 , while the subsequent n bits, A_2 , are added to the first branch P_1 . Afterwards, the first branch is encrypted and XORed to the second branch. Then, the second branch is encrypted, and the outputs of the encryptions are XORed to generate the seed, J , for the counter mode operation. The counter mode produces a keystream which is used to encrypt the third plaintext segment $\overline{P}$, yielding the ciphertext segment $\overline{C}$. Since the seed J for the counter mode incorporates P_1 and P_2 , the generated keystream ensures that both initial plaintext segments influence the ciphertext $\overline{C}$ of the third segment. Up to this stage, the first branch, P_1 , has not been directly influenced by the second branch, P_2 . To rectify this, the hash of the ciphertext segment, $\overline{C}$, which implicitly contains information about both P_1 and P_2 , is XORed back to these two branches in accordance with their original respective sizes. Following this, in a departure from the standard hash-encrypt-hash method, the modified second branch is encrypted using the block cipher, E . This encrypted output is then XORed with the first branch, and the modified first branch is also encrypted. Finally, the complete ciphertext, C , is constructed by the concatenation of the encrypted first segment, C_1 , the encrypted second segment, C_2 , and the ciphertext of the third segment, $\overline{C}$. This results in the final structure: $C = C_1 \parallel C_2 \parallel \overline{C}$.

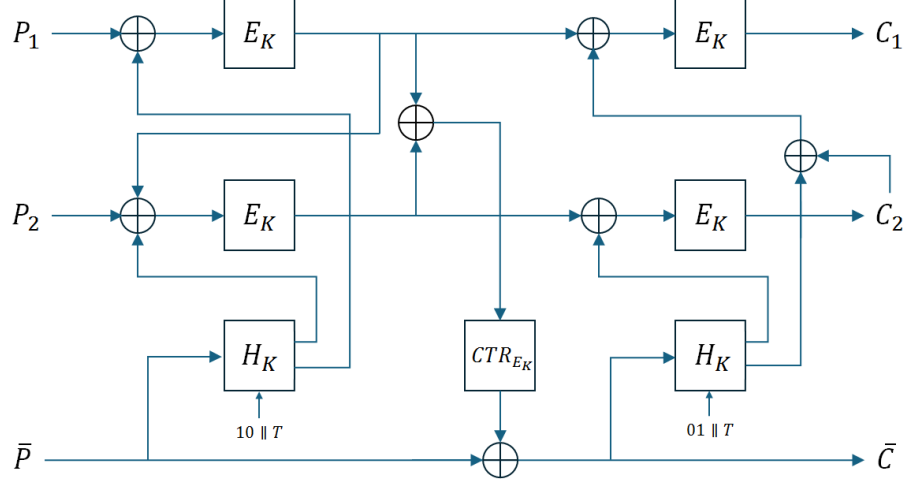


Figure 1: Animagus Encryption Algorithm

Algorithm 1 $\text{Enc}(K, T, P)$	Algorithm 2 $\text{Dec}(K, T, C)$
$P_1 \leftarrow \text{MSB}_n(P)$	$C_1 \leftarrow \text{MSB}_n(C)$
$P_2 \leftarrow \text{MSB}_n(P \ll n)$	$C_2 \leftarrow \text{MSB}_n(C \ll n)$
$\bar{P} \leftarrow \text{LSB}_{ P -2n}(P)$	$\bar{C} \leftarrow \text{LSB}_{ C -2n}(C)$
$X_1 \leftarrow H_K(\bar{P} \parallel 10 \parallel T)$	$\bar{X}_1 \leftarrow H_K(\bar{C} \parallel 01 \parallel T)$
$A_1 \leftarrow \text{MSB}_n(X_1)$	$\bar{A}_1 \leftarrow \text{MSB}_n(\bar{X}_1)$
$A_2 \leftarrow \text{LSB}_n(X_1)$	$\bar{A}_2 \leftarrow \text{LSB}_n(\bar{X}_1)$
$F \leftarrow E_K(P_1 \oplus A_2)$	$\bar{F} \leftarrow D_K(C_1) \oplus \bar{A}_2 \oplus C_2$
$B \leftarrow E_K(P_2 \oplus A_1 \oplus F)$	$\bar{B} \leftarrow D_K(C_2) \oplus \bar{A}_1$
$J \leftarrow F \oplus B$	$\bar{J} \leftarrow \bar{F} \oplus \bar{B}$
$L \leftarrow \text{CTR}_{E_K}(J)$	$\bar{L} \leftarrow \text{CTR}_{E_K}(\bar{J})$
$\bar{C} \leftarrow \bar{P} \oplus L$	$\bar{P} \leftarrow \bar{C} \oplus \bar{L}$
$X_2 \leftarrow H_K(\bar{C} \parallel 01 \parallel T)$	$\bar{X}_2 \leftarrow H_K(\bar{P} \parallel 10 \parallel T)$
$A_3 \leftarrow \text{MSB}_n(X_2)$	$\bar{A}_3 \leftarrow \text{MSB}_n(\bar{X}_2)$
$A_4 \leftarrow \text{LSB}_n(X_2)$	$\bar{A}_4 \leftarrow \text{LSB}_n(\bar{X}_2)$
$C_2 \leftarrow E_K(B \oplus A_3)$	$P_2 \leftarrow D_K(\bar{B}) \oplus \bar{A}_3 \oplus \bar{F}$
$C_1 \leftarrow E_K(F \oplus A_4 \oplus C_2)$	$P_1 \leftarrow D_K(\bar{F}) \oplus \bar{A}_4$
$C \leftarrow C_1 \parallel C_2 \parallel \bar{C}$	$P \leftarrow P_1 \parallel P_2 \parallel \bar{P}$
return C	return P

3.1 Special Accordion Parameters

As evident in the general accordion design principles, all operational parameters are essentially linked to the characteristics of the underlying block cipher employed within the accordion mode. Consequently, if a block cipher with a block size of $n = 256$ bits is utilized, the key size of the accordion mode directly corresponds to the key size of this block cipher. Furthermore, the message length should fall within the range of 512 bits to 2^{120} bits, and the HMAC function must employ SHA3-512 as its hash primitive. Conversely, when the AES block cipher is selected, the block size n is fixed at 128 bits, while the key size offers three distinct options: 128 bits, 192 bits, and 256 bits. In this specific scenario, the acceptable message length is constrained between 256 bits and 2^{55} bits, and SHA3-256 must be used in conjunction with HMAC. Notably, due to the construction of the tweak-based term T within the accordion framework, any instantiation of a special accordion mode possesses an effectively unlimited tweak size.

3.2 Different Accordion Applications

NIST requests the realization of three distinct applications through the utilization of derived functions: Authenticated Encryption with Associated Data, Tweakable Encryption, and Deterministic Authenticated Encryption. The provision of the requisite properties for each of these applications necessitates specific architectural designs within the derived function framework.

3.2.1 Authenticated Encryption with Associated Data

An Authenticated Encryption with Associated Data (AEAD) scheme is defined as an encryption algorithm that concurrently provides confidentiality for the message and integrity for both the message and associated data. Our construction achieves message authentication by embedding fixed-value bits into the plaintext and leveraging the accordion tweak to incorporate the nonce and associated data as the recommendations outlined in [6].

Specifically, a fixed n -bit value, designated as the authentication data (AUD), is prepended to the plaintext. The full plaintext for encryption is structured as $AUD || P_1 || \bar{P}$, where P_1 represents the first n bits of the message and $\bar{P}$ is the remainder. During decryption, the first n bits of the output are extracted and compared against the original fixed n -bit value to verify integrity.

The accordion tweak is computed as the output of a keyed hash function, HMAC-SHA3-128, applied to the concatenation of the nonce (N) and the

associated data (AD). This operation utilizes the same key (K) as the underlying accordion mode, defined as $T = \text{HMAC}_K(N||AD)$.

3.2.2 Tweakable Encryption

A tweakable encryption scheme is fundamentally an encryption algorithm that accepts an additional input parameter known as the tweak. For this particular application, given the emphasis on message confidentiality without the requirement for inherent authentication, there are no dedicated authentication bits within the plaintext structure. As the underlying accordion mode design inherently incorporates a tweak input, we can effectively encode supplementary information by employing HMAC with SHA3-128 to generate the tweak value, which is then provided as input to the accordion mode. Modifying the tweak necessitates only a single hash computation, thus maintaining a low level of operational complexity.

3.2.3 Deterministic Authenticated Encryption

A Deterministic Authenticated Encryption (DAE) scheme is characterized as an encryption algorithm that provides both confidentiality and message authentication while operating deterministically. This implies the absence of any random values or, in this context, a tweak input. Our proposed accordion mode can be adapted for DAE by simply omitting the tweak input without loss of functionality.

The requisite message authentication for DAE is achieved by incorporating a fixed n -bit authentication value AUD at the beginning of the plaintext, mirroring the approach used for the AEAD construction. The plaintext is thus structured as $AUD||P_1||\bar{P}$, where P_1 is the first n bits of the message and $\bar{P}$ is the remainder. Post-decryption, the first n bits of the output are verified against the predefined AUD value to confirm integrity.

4 Security Analysis of the Design

The security definition employed in the formal proof of the presented Animagus algorithm is adapted from [19]. Let $\mathcal{M}$ denote the message space and $\mathcal{K}$ represent the key space. Consider $E : \mathcal{K} \times \mathcal{M} \rightarrow \mathcal{M}$ as the encryption algorithm under analysis, and $\pi : \mathcal{M} \rightarrow \mathcal{M}$ as a length-preserving permutation, where $\text{Perm}(\mathcal{M})$ signifies the set of all such permutations. For an adversary $\mathcal{A}$ that attempts to distinguish the encryption algorithm E from

a randomly chosen permutation, the advantage of $\mathcal{A}$ in successfully winning this distinguishing game is quantified as:

$$\mathbf{Adv}_E^{SPRP} := |Pr[K \xleftarrow{\$} \mathcal{K} : \mathcal{A}^{E_K, E_K^{-1}} \implies 1] - Pr[\pi \xleftarrow{\$} \text{Perm}(\mathcal{M}) : \mathcal{A}^{\pi, \pi^{-1}} \implies 1]|$$

This advantage measures the adversary's ability to differentiate between outputs produced by the algorithm E under a randomly chosen key K and outputs generated by a truly random permutation π over the message space $\mathcal{M}$. A negligible advantage indicates that the algorithm E exhibits strong pseudorandom permutation security.

The notation $\mathcal{A} \implies 1$ signifies that the adversary outputs 1 in the SPRP security game, where 0 denotes that the adversary has the encryption algorithm oracle E as a challenger, and 1 indicates that the adversary has the random permutation oracle π as a challenger, based on a random choice made at the beginning of the game. The explicit inclusion of both the forward (E, π) and inverse (E^{-1}, π^{-1}) functionalities in the adversary's oracle access $\mathcal{A}^{E_K, E_K^{-1}}$ and $\mathcal{A}^{\pi, \pi^{-1}}$ signifies that the security game encompasses both chosen ciphertext attacks and chosen plaintext attacks. This comprehensive adversarial capability is precisely what defines the notion of a strong pseudorandom permutation. A secure encryption scheme under this definition implies that no computationally feasible adversary can exhibit a non-negligible advantage in distinguishing the encryption algorithm from a truly random permutation, even when provided with the ability to adaptively query both the encryption and decryption functionalities.

The security of the proposed Animagus accordion mode, as defined by the strong pseudorandom permutation notion elaborated above, is formally stated in the subsequent theorem. This theorem establishes an upper bound on the adversarial advantage against Animagus in terms of the adversarial advantage against the underlying block cipher from which Animagus is constructed. This bounding relationship provides a measure of the security of Animagus relative to the assumed security of its fundamental building block.

Theorem 1. *Let $E : \mathcal{K} \times \{0, 1\}^n \rightarrow \{0, 1\}^n$ be a block cipher, and H be the HMAC-SHA3 hash function. Let $\text{Animagus}[E, H]$ be the accordion mode construction with message space $\mathcal{M} = \{0, 1\}^{na}$ and key space $\mathcal{K} = \{0, 1\}^k$. Let $\mathcal{A}$ be an adversary that makes a total of q queries to the oracle, including both encryption and decryption requests. The SPRP advantage of any such adversary is bounded as follows:*

$$\mathbf{Adv}_{\text{Animagus}[E, H]}^{SPRP} \leq 2 \cdot \mathbf{Adv}_E^{SPRP} + \frac{\binom{q(a+2)}{2}}{2^{n-1}} + \frac{\binom{q}{2}}{2^n} + \frac{q^2}{2^{n+1}} + \frac{\binom{q}{2}}{2^{2n}}$$

where the adversary makes q encryption/decryption queries.

To rigorously establish the security of the Animagus construction as articulated in the preceding theorem, we employ the game-substitution argument, a proof technique notably applied in the seminal work by Halevi and Rogaway [19]. The core strategy of this method involves a sequence of game transformations, where each step aims to incrementally simplify the adversary’s interaction with the system. The ultimate objective is to arrive at a final, simplified game scenario, ideally a non-interactive setting devoid of any effective adversarial advantage. By meticulously demonstrating that the adversary’s probability of success changes negligibly across each game transition, we can ultimately bound the adversary’s advantage in the original security game by the advantage in the final, trivial game, thereby proving the SPRP security of the Animagus algorithm relative to the security of its underlying components. We adhere to a similar sequence of game transformations as presented in [19] to systematically demonstrate the SPRP security of our proposed construction.

Proof. Let q denote the number of queries made by an adversary $\mathcal{A}$ to a given oracle. Each query has a bit length of $n \cdot a$, where n is the block size of the underlying encryption algorithm and a is the number of blocks. As previously mentioned, the proof of the theorem proceeds via a hybrid argument, a sequence of games starting with the real-world algorithm and culminating in an ideal, random oracle.

Initially, we define **Game0** as the actual Animagus algorithm. We then transition to **Game1** by replacing the block cipher with a random permutation. This change is valid because the adversary’s advantage in distinguishing between the two games is bounded by the security of the block cipher itself.

Next, we progress to **Game2**, where the hash algorithm is substituted with a random function. This transition introduces an additional term to the security bound, specifically the collision probability of the hash function. The cumulative advantage is now limited by the sum of the block cipher’s distinguishing probability and the hash function’s collision probability.

Having randomized the core components of the Animagus algorithm, the subsequent game transitions will lead to a final state where the adversary is interacting with a completely random oracle. The total distinguishing advantage of the adversary is then bounded by the sum of the advantages from each game transition.

Game0 : In this game, the adversary is provided with an oracle that provides responses from the $\text{Animagus}[E, H]$ construction. Here, E is a block

cipher with a block size of n bits, and H is the HMAC-SHA hash algorithm, which produces a $2n$ -bit output. Therefore, the adversary's probability of winning this game is identical to the probability of winning **Game0**.

$$Pr[\mathcal{A}^{\text{Animagus}[E,H]} \Rightarrow 1] = Pr[\mathcal{A}^{\text{Game0}} \Rightarrow 1]$$

Game1 : In this game, we change E with a random permutation π and construct $\text{Animagus}[P, H]$ variant. Therefore, it provides an oracle to the adversary $\mathcal{A}$ that emulates the behavior of $\text{Animagus}[P, H]$. Consequently, we subtract the distinguishing advantage of the block cipher from the adversary's advantage, yielding the following inequality:

$$\mathbf{Adv}_{\text{Animagus}[E,H]}^{\text{SPRP}} \leq \mathbf{Adv}_{\text{Animagus}[P,H]}^{\text{SPRP}} + 2 \cdot \mathbf{Adv}_E^{\text{SPRP}}$$

The permutation π is adaptively constructed throughout the game in response to $\mathcal{A}$'s queries and the corresponding answers. Initially, π is undefined for all inputs. When the oracle receives a query requiring the evaluation of $\pi(X)$ for some input X , the following occurs: if $\pi(X)$ is currently undefined, a uniformly random value Y is selected from the codomain, and the mapping $\pi(X) = Y$ is established. Similarly, if a query necessitates the evaluation of the inverse permutation $\pi^{-1}(Y)$ for some output Y , and there is no existing X such that $\pi(X) = Y$, a uniformly random value X from the domain is chosen, and the mapping $\pi(X) = Y$ is set. In both cases, the newly defined input X and output Y are added to the domain (D) and range (R) of π , respectively. Upon receiving any query, the oracle first checks if the required input or output is already within the defined domain or range of π to ensure the consistency and well-defined nature of the permutation. Let $\bar{D}$ and $\bar{R}$ denote the complements of the domain and range, respectively.

A flag, denoted as *bad*, is set to true if and only if a value that has been previously chosen is selected again. This flag is solely for the purpose of the security analysis and is not accessible or observable by the adversary $\mathcal{A}$.

Game2 : In this game, the cryptographic hash function H , specifically HMAC-SHA3, is replaced by a truly random function F , leading to a new construction, $\text{Animagus}[P, F]$ 3, 4. This substitution is justified by treating SHA3 as an ideal random oracle. It is assumed to produce independent, uniformly random outputs for distinct inputs. The analysis proceeds under the assumption that the adversary submits distinct queries, as redundant queries provide no new information and can be disregarded without loss of generality.

Consequently, the adversary's advantage in distinguishing the game with the real hash function ($\text{Animagus}[P, H]$) from the game with the random

function ($\text{Animagus}[P, F]$) is bounded by the probability of a hash collision. For q queries and a $2n$ -bit output, this collision probability is approximately $\binom{q}{2} \cdot 2^{-2n}$. This relationship is formally expressed by the following inequality, which bounds the SPRP advantage:

$$\mathbf{Adv}_{\text{Animagus}[P,H]}^{\text{SPRP}} \leq \mathbf{Adv}_{\text{Animagus}[P,F]}^{\text{SPRP}} + \frac{\binom{q}{2}}{2^{2n}}$$

This hybrid argument demonstrates that the difference in security between the two game scenarios is negligible, as it is limited by the birthday bound on the hash function's output space.

Before we continue, to simplify the security analysis, we transition from bounding the SPRP advantage to bounding the Strong Pseudorandom Function (SPRF) advantage. This is a common and effective proof technique.

The motivation for this transition is that the SPRF security game is often simpler to analyze. It does not require considering the complexity of inverse queries (decryption), which are a fundamental part of the SPRP game. Therefore, we can use a well-known relationship to bound the SPRP advantage, which states that the SPRP advantage can be upper-bounded by the SPRF advantage plus the probability of the function not being a permutation. This reduces the number of *bad* events that must be considered and bounded throughout the game-hopping sequence. This relationship between SPRP-security and SPRF-security, as detailed in [1] is the following:

$$\mathbf{Adv}_{\text{Animagus}[P,H]}^{\text{SPRP}} \leq \mathbf{Adv}_{\text{Animagus}[P,H]}^{\text{SPRF}} + \frac{q^2}{2^{n+1}}$$

Using the relations from the previous games and above inequality, we can bound the total SPRP advantage of the Animagus algorithm as the following:

$$\begin{aligned} \mathbf{Adv}_{\text{Animagus}[E,H]}^{\text{SPRP}} &\leq \mathbf{Adv}_{\text{Animagus}[P,H]}^{\text{SPRP}} + 2 \cdot \mathbf{Adv}_E^{\text{SPRP}} \\ &\leq \mathbf{Adv}_{\text{Animagus}[P,F]}^{\text{SPRP}} + 2 \cdot \mathbf{Adv}_E^{\text{SPRP}} + \frac{\binom{q}{2}}{2^{2n}} \\ &\leq \mathbf{Adv}_{\text{Animagus}[P,F]}^{\text{SPRF}} + 2 \cdot \mathbf{Adv}_E^{\text{SPRP}} + \frac{q^2}{2^{n+1}} + \frac{\binom{q}{2}}{2^{2n}} \end{aligned}$$

Game3 : This game is derived from **Game2** by removing the mechanisms that ensure the consistency of π , which correspond to the boxed statements in the description of **Game2**. This modification implies that previously established mappings can be altered with subsequent choices. As the consistency checks associated with its definition are no longer present, the permutation π is also entirely removed.

Algorithm 3 Game2 - r-th Encryption Query($P_1^r || P_2^r || \overline{P}^r$)

$bad \leftarrow \text{false}; \quad D \leftarrow R \leftarrow \emptyset; \quad \pi \leftarrow \text{undef}$
 $X_1^r \xleftarrow{\$} \{0, 1\}^{2n}$
 $A_1^r \leftarrow \text{MSB}_n(X_1^r), \quad A_2^r \leftarrow \text{LSB}_n(X_1^r)$
 $F^r \xleftarrow{\$} \{0, 1\}^n$
if $F^r \in R$ **then** $bad \leftarrow \text{true}, \boxed{F^r \xleftarrow{\$} \overline{R}}$
if $(P_1^r \oplus A_2^r) \in D$ **then** $bad \leftarrow \text{true}, \boxed{F^r \leftarrow \pi(P_1^r \oplus A_2^r)}$
 $\pi(P_1^r \oplus A_2^r) \leftarrow F^r, \quad D \leftarrow D \cup \{P_1^r \oplus A_2^r\}, \quad R \leftarrow R \cup \{F^r\}$
 $B^r \xleftarrow{\$} \{0, 1\}^n$
if $B^r \in R$ **then** $bad \leftarrow \text{true}, \boxed{B^r \xleftarrow{\$} \overline{R}}$
if $(P_2^r \oplus A_1^r \oplus F^r) \in D$ **then** $bad \leftarrow \text{true}, \boxed{B^r \leftarrow \pi(P_2^r \oplus A_1^r \oplus F^r)}$
 $\pi(P_2^r \oplus A_1^r \oplus F^r) \leftarrow B^r, \quad D \leftarrow D \cup \{P_2^r \oplus A_1^r \oplus F^r\}, \quad R \leftarrow R \cup \{B^r\}$
 $J^r \leftarrow F^r \oplus B^r$
 $L^r \leftarrow \text{CTR}_\pi(J^r)$
 $L^r = L_0^r \cdots L_b^r$ where $b = \lfloor |\overline{P}^r|/n \rfloor$
for $i = 0$ **to** b **do**
 $L_i^r \xleftarrow{\$} \{0, 1\}^n$
if $L_i^r \in R$ **then** $bad \leftarrow \text{true}, \boxed{L_i^r \xleftarrow{\$} \overline{R}}$
if $(J^r \oplus i) \in D$ **then** $bad \leftarrow \text{true}, \boxed{L_i^r \leftarrow \pi(J^r \oplus i)}$
 $\pi(J^r \oplus i) \leftarrow L_i^r, \quad D \leftarrow D \cup \{J^r \oplus i\}, \quad R \leftarrow R \cup \{L_i^r\}$
end for
 $\overline{C}^r \leftarrow \overline{P}^r \oplus L^r$
 $X_2^r \xleftarrow{\$} \{0, 1\}^{2n}$
 $A_3^r \leftarrow \text{MSB}_n(X_2^r), \quad A_4^r \leftarrow \text{LSB}_n(X_2^r)$
 $C_2^r \xleftarrow{\$} \{0, 1\}^n$
if $C_2^r \in R$ **then** $bad \leftarrow \text{true}, \boxed{C_2^r \xleftarrow{\$} \overline{R}}$
if $(B^r \oplus A_3^r) \in D$ **then** $bad \leftarrow \text{true}, \boxed{C_2^r \leftarrow \pi(B^r \oplus A_3^r)}$
 $\pi(B^r \oplus A_3^r) \leftarrow C_2^r, \quad D \leftarrow D \cup \{B^r \oplus A_3^r\}, \quad R \leftarrow R \cup \{C_2^r\}$
 $C_1^r \xleftarrow{\$} \{0, 1\}^n$
if $C_1^r \in R$ **then** $bad \leftarrow \text{true}, \boxed{C_1^r \xleftarrow{\$} \overline{R}}$
if $(F^r \oplus A_4^r \oplus C_2^r) \in D$ **then** $bad \leftarrow \text{true}, \boxed{C_1^r \leftarrow \pi(F^r \oplus A_4^r \oplus C_2^r)}$
 $\pi(F^r \oplus A_4^r \oplus C_2^r) \leftarrow C_1^r, \quad D \leftarrow D \cup \{F^r \oplus A_4^r \oplus C_2^r\}, \quad R \leftarrow R \cup \{C_1^r\}$
 $C^r \leftarrow C_1^r || C_2^r || \overline{C}^r$
return C^r

Algorithm 4 Game2 - r-th Decryption Query($C_1^r || C_2^r || \overline{C}^r$)

$bad \leftarrow \mathbf{false}; \quad D \leftarrow R \leftarrow \emptyset; \quad \pi \leftarrow \mathbf{undef}$
 $\overline{X}_1^r \xleftarrow{\$} \{0, 1\}^{2n}$
 $\overline{A}_1^r \leftarrow \text{MSB}_n(\overline{X}_1^r), \quad \overline{A}_2^r \leftarrow \text{LSB}_n(\overline{X}_1^r)$
 $\overline{B}^r \xleftarrow{\$} \{0, 1\}^n$
if $C_2^r \in R$ **then** $bad \leftarrow \mathbf{true}, \boxed{C_2^r \xleftarrow{\$} \overline{R}}$
if $(\overline{B}^r \oplus \overline{A}_1^r) \in D$ **then** $bad \leftarrow \mathbf{true}, \boxed{C_2^r \leftarrow \pi(\overline{B}^r \oplus \overline{A}_1^r)}$
 $\pi(\overline{B}^r \oplus \overline{A}_1^r) \leftarrow C_2^r, \quad D \leftarrow D \cup \{\overline{B}^r \oplus \overline{A}_1^r\}, \quad R \leftarrow R \cup \{C_2^r\}$
 $\overline{F}^r \xleftarrow{\$} \{0, 1\}^n$
if $C_1^r \in R$ **then** $bad \leftarrow \mathbf{true}, \boxed{C_1^r \xleftarrow{\$} \overline{R}}$
if $(\overline{F}^r \oplus \overline{A}_2^r \oplus C_2^r) \in D$ **then** $bad \leftarrow \mathbf{true}, \boxed{C_1^r \leftarrow \pi(\overline{F}^r \oplus \overline{A}_2^r \oplus C_2^r)}$
 $\pi(\overline{F}^r \oplus \overline{A}_2^r \oplus C_2^r) \leftarrow C_1^r, \quad D \leftarrow D \cup \{\overline{F}^r \oplus \overline{A}_2^r \oplus C_2^r\}, \quad R \leftarrow R \cup \{C_1^r\}$
 $\overline{J}^r = \overline{F}^r \oplus \overline{B}^r$
 $\overline{L}^r \leftarrow \text{CTR}_\pi(\overline{J}^r)$
 $\overline{L}^r = \overline{L}_0^r \cdots \overline{L}_b^r \text{ where } b = \lfloor |\overline{C}|/n \rfloor$
for $i = 0$ **to** b **do**
 $\overline{L}_i^r \xleftarrow{\$} \{0, 1\}^n$
if $\overline{L}_i^r \in R$ **then** $bad \leftarrow \mathbf{true}, \boxed{\overline{L}_i^r \xleftarrow{\$} \overline{R}}$
if $(\overline{J}^r \oplus i) \in D$ **then** $bad \leftarrow \mathbf{true}, \boxed{\overline{L}_i^r \leftarrow \pi(\overline{J}^r \oplus i)}$
 $\pi(\overline{J}^r \oplus i) \leftarrow \overline{L}_i^r, \quad D \leftarrow D \cup \{\overline{J}^r \oplus i\}, \quad R \leftarrow R \cup \{\overline{L}_i^r\}$
end for
 $\overline{P}^r \leftarrow \overline{C}^r \oplus \overline{L}^r$
 $\overline{X}_2^r \xleftarrow{\$} \{0, 1\}^{2n}$
 $\overline{A}_3^r \leftarrow \text{MSB}_n(\overline{X}_2^r), \quad \overline{A}_4^r \leftarrow \text{LSB}_n(\overline{X}_2^r)$
 $\overline{P}_2^r \xleftarrow{\$} \{0, 1\}^n$
if $\overline{B}^r \in R$ **then** $bad \leftarrow \mathbf{true}, \boxed{\overline{B}^r \xleftarrow{\$} \overline{R}}$
if $(\overline{P}_2^r \oplus \overline{A}_3^r \oplus \overline{F}^r) \in D$ **then** $bad \leftarrow \mathbf{true}, \boxed{\overline{B}^r \leftarrow \pi(\overline{P}_2^r \oplus \overline{A}_3^r \oplus \overline{F}^r)}$
 $\pi(\overline{P}_2^r \oplus \overline{A}_3^r \oplus \overline{F}^r) \leftarrow \overline{B}^r, \quad D \leftarrow D \cup \{\overline{P}_2^r \oplus \overline{A}_3^r \oplus \overline{F}^r\}, \quad R \leftarrow R \cup \{\overline{B}^r\}$
 $\overline{P}_1^r \xleftarrow{\$} \{0, 1\}^n$
if $\overline{F}^r \in R$ **then** $bad \leftarrow \mathbf{true}, \boxed{\overline{F}^r \xleftarrow{\$} \overline{R}}$
if $(\overline{P}_1^r \oplus \overline{A}_4^r) \in D$ **then** $bad \leftarrow \mathbf{true}, \boxed{\overline{F}^r \leftarrow \pi(\overline{P}_1^r \oplus \overline{A}_4^r)}$
 $\pi(\overline{P}_1^r \oplus \overline{A}_4^r) \leftarrow \overline{F}^r, \quad D \leftarrow D \cup \{\overline{P}_1^r \oplus \overline{A}_4^r\}, \quad R \leftarrow R \cup \{\overline{F}^r\}$
 $\overline{P}^r \leftarrow \overline{P}_1^r || \overline{P}_2^r || \overline{P}^r$
return $\overline{P}^r$

However, the probability of the *bad* flag being set to true remains identical in both games, as the conditions under which *bad* is set have not been changed. Given that the sole difference between **Game2** and **Game3** lies in the consistency checks for the permutation, we can conclude that:

$$Pr[\mathcal{A}^{\text{Game2}} \implies 1] - Pr[\mathcal{A}^{\text{Game3}} \implies 1] \leq Pr[\text{Game3 sets } bad]$$

Another change is that the random values L and $\bar{L}$, which were initially chosen and then used to compute $\bar{C}$ (in encryption queries) or $\bar{P}$ (in decryption queries) in **Game2**, are now directly chosen as $\bar{C}$ or $\bar{P}$ in **Game3**. Specifically, in **Game2**, for an encryption query, L is randomly selected, and then $\bar{C} = \bar{P} \oplus L$ is computed. Similarly, for a decryption query in **Game2**, $\bar{L}$ is randomly chosen, and then $\bar{P} = \bar{C} \oplus \bar{L}$ is computed. In **Game3**, for an encryption query, $\bar{C}$ is directly chosen as random, and then $L = \bar{P} \oplus \bar{C}$ is computed. For a decryption query in **Game3**, $\bar{P}$ is directly chosen as random, and then $\bar{L} = \bar{C} \oplus \bar{P}$ is computed. These changes do not affect the adversary's view of the oracle's responses.

Now, considering **Game3**, we observe that for an encryption query, $C_1 || C_2 || \bar{C}$ are chosen randomly, and for a decryption query, $P_1 || P_2 || \bar{P}$ are chosen randomly. This implies that the adversary receives completely random bits as responses. Consequently, **Game3** behaves like a random bit oracle from the adversary's perspective:

$$Pr[\mathcal{A}^{\text{Game3}} \implies 1] = Pr[\mathcal{A}^{\text{SPRF}} \implies 1]$$

By combining the distinguishing advantages from each game transition with the formal definition of SPRF security, we can derive:

$$\begin{aligned} \text{Adv}_{\text{Animagus}[P,F]}^{\text{SPRF}} &= |Pr[\mathcal{A}^{\text{Animagus}[P,F]} \implies 1] - Pr[\mathcal{A}^{\text{SPRF}} \implies 1]| \\ &= |Pr[\mathcal{A}^{\text{Game2}} \implies 1] - Pr[\mathcal{A}^{\text{Game3}} \implies 1]| \\ &\leq Pr[\text{Game3 sets } bad] \end{aligned}$$

Based on these inequalities, bounding the probability of the *bad* event occurring in **Game3** will be sufficient to bound the SPRF advantage of the adversary.

Game4 : In this game, we restructure the process into three distinct steps. First, for each of the adversary's queries, we directly return responses that are chosen uniformly at random. Second, after all queries from the adversary have been processed, we define the intermediate variables that would have been computed during the interaction for all the queries made. Finally, based on the values determined in the second step, we examine the sets D (domain)

and R (range). The *bad* flag is set to true if any value appears more than once within at least one of these sets. **Game4** is described in Algorithm 5. Although the order of operations is altered, the underlying conditions that lead to the setting of the *bad* flag remain unchanged. Consequently, the probability of the *bad* event occurring is the same as in **Game3**:

$$Pr[\text{Game3 sets bad}] = Pr[\text{Game4 sets bad}]$$

Game5 : In this game, we eliminate the adversary and the interactive aspect. The queries and their corresponding responses no longer influence the probability of the *bad* flag being set. However, under certain conditions, *bad* is set to true with probability 1. As we mentioned before, if the adversary were to repeat the same query, it would gain no new information, but the *bad* flag would be definitively set. These pointless queries are assumed to be excluded in these games. Similarly, there are implicit constraints on the generated answers to avoid directly triggering the *bad* condition. Based on the algorithm's structure, these restrictions for two distinct queries, the r -th and the s -th, are as follows:

- $C_1^r \neq C_1^s$ and $C_2^r \neq C_2^s$ since $R \leftarrow R \cup \{C_1\}$ and $R \leftarrow R \cup \{C_2\}$ in encryption query
- $\overline{C}^r \oplus \overline{P}^r \neq \overline{C}^s \oplus \overline{P}^s$ since $L \leftarrow \overline{C} \oplus \overline{P}$, $R \leftarrow R \cup \{L\}$ in encryption query
- $\overline{P}^r \oplus \overline{C}^r \neq \overline{P}^s \oplus \overline{C}^s$ since $\overline{L} \leftarrow \overline{P} \oplus \overline{C}$, $R \leftarrow R \cup \{\overline{L}\}$ in decryption query

The probability of a collision occurring within the range R , given at most q elements from q queries, is $\binom{q}{2} \cdot 2^{-n}$. This represents the probability that at least two of the chosen values are identical. **Game5** considers a fixed set of queries and answers that adhere to these restrictions, maximizing the probability of the *bad* flag being set. Therefore, the relationship between **Game4** and **Game5** can be expressed as:

$$Pr[\text{Game4 sets bad}] \leq Pr[\text{Game5 sets bad}] + \frac{\binom{q}{2}}{2^n}$$

Game6: To facilitate the analysis, we simplify the game by disregarding the parts related to the range set R . Because, we have an inherent symmetry which means that the domain and range sets having elements with same probability distribution. Due to this inherent symmetry between the domain and the range in the construction, we can focus solely on the domain D without loss of generality, thereby reducing the number of cases to examine. This game presented in Algorithm 6. Exploiting this symmetry, we

Algorithm 5 Game4

RESPONDING QUERIES

r-th Encryption Query ($P_1^r || P_2^r || \overline{P}^r$)

$qt^r \leftarrow \mathbf{Enc}$

$CC^r \xleftarrow{\$} \{0, 1\}^{an}, l = |P_1^r || P_2^r || \overline{P}^r|$

$C^r = C_1^r || C_2^r || \overline{C}^r \leftarrow \text{MSB}_l(CC^r)$

return C^r

r-th Decryption Query ($C_1^r || C_2^r || \overline{C}^r$)

$qt^r \leftarrow \mathbf{Dec}$

$PP^r \xleftarrow{\$} \{0, 1\}^{an}, l = |C_1^r || C_2^r || \overline{C}^r|$

$P^r = P_1^r || P_2^r || \overline{P}^r \xleftarrow{\$} \text{MSB}_l(PP^r)$

return P^r

DEFINING INTERMEDIATE VARIABLES

$D \leftarrow R \leftarrow \emptyset$, where D, R are multisets

for $r = 1$ to q **do**

if $qt^r = \mathbf{Enc}$ **then**

$X_1^r \xleftarrow{\$} \{0, 1\}^{2n}$

$A_1^r \leftarrow \text{MSB}_n(X_1^r), A_2^r \leftarrow \text{LSB}_n(X_1^r)$

$F^r \xleftarrow{\$} \{0, 1\}^n; D \leftarrow D \cup \{P_1^r \oplus A_2^r\}, R \leftarrow R \cup \{F^r\}$

$B^r \xleftarrow{\$} \{0, 1\}^n; D \leftarrow D \cup \{P_2^r \oplus A_1^r \oplus F^r\}, R \leftarrow R \cup \{B^r\}$

$J^r \leftarrow F^r \oplus B^r$

$L^r = L_0^r \cdots L_b^r \leftarrow \overline{C}^r \oplus \overline{P}^r$, where $b = \lfloor |\overline{P}|/n \rfloor$

for $i = 0$ to b **do**

$D \leftarrow D \cup \{J^r \oplus i\}, R \leftarrow R \cup \{L_i^r\}$

end for

$X_2^r \xleftarrow{\$} \{0, 1\}^{2n}$

$A_3^r \leftarrow \text{MSB}_n(X_2^r), A_4^r \leftarrow \text{LSB}_n(X_2^r)$

$D \leftarrow D \cup \{B^r \oplus A_3^r\}, R \leftarrow R \cup \{C_2^r\}$

$D \leftarrow D \cup \{F^r \oplus A_4^r \oplus C_2^r\}, R \leftarrow R \cup \{C_1^r\}$

else if $qt^r = \mathbf{Dec}$ **then**

$\overline{X}_1^r \xleftarrow{\$} \{0, 1\}^{2n}$

$\overline{A}_1^r \leftarrow \text{MSB}_n(\overline{X}_1^r), \overline{A}_2^r \leftarrow \text{LSB}_n(\overline{X}_1^r)$

$\overline{B}^r \xleftarrow{\$} \{0, 1\}^n; D \leftarrow D \cup \{\overline{B}^r \oplus \overline{A}_1^r\}, R \leftarrow R \cup \{C_2^r\}$

$\overline{F}^r \xleftarrow{\$} \{0, 1\}^n; D \leftarrow D \cup \{\overline{F}^r \oplus \overline{A}_2^r \oplus C_2^r\}, R \leftarrow R \cup \{C_1^r\}$

$J^r \leftarrow F^r \oplus B^r$

$\overline{L}^r = \overline{L}_0^r \cdots \overline{L}_b^r \leftarrow \overline{P}^r \oplus \overline{C}^r$, where $b = \lfloor |\overline{C}|/n \rfloor$

for $i = 0$ to b **do**

$D \leftarrow D \cup \{\overline{J}^r \oplus i\}, R \leftarrow R \cup \{\overline{L}_i^r\}$

end for

$\overline{X}_2^r \xleftarrow{\$} \{0, 1\}^{2n}$

$\overline{A}_3^r \leftarrow \text{MSB}_n(\overline{X}_2^r), \overline{A}_4^r \leftarrow \text{LSB}_n(\overline{X}_2^r)$

$D \leftarrow D \cup \{P_2^r \oplus \overline{A}_3^r \oplus \overline{F}^r\}, R \leftarrow R \cup \{\overline{B}^r\}$

$D \leftarrow D \cup \{P_1^r \oplus \overline{A}_4^r\}, R \leftarrow R \cup \{\overline{F}^r\}$

end if

end for

DEFINING VARIABLE *bad*

bad $\leftarrow$ true **if** (some value appears more than once in D or R)

can assume that the probability of a value appearing more than once in the domain is greater than or equal to the probability of a value appearing more than once in the range: $\Pr[\text{some value appears more than once in } D] \geq \Pr[\text{some value appears more than once in } R]$.

From this, we can establish the bound: $\Pr[\text{Game5 sets bad}] \leq 2 \cdot \Pr[\text{Game6 sets bad}]$. Consequently, to prove the theorem, we now focus on bounding $\Pr[\text{Game6 sets bad}]$. By combining this with the inequalities derived from the previous games, we obtain:

$$\mathbf{Adv}_{\text{Animagus}[P,F]}^{\text{SPRF}} \leq 2 \cdot \Pr[\text{Game6 sets bad}] + \frac{\binom{q}{2}}{2^n}$$

Therefore, bounding $\Pr[\text{Game6 sets bad}]$ will allow us to establish the desired bound on the SPRF advantage.

Algorithm 6 Game6

$Q = (qt, P_1 || P_2 || \overline{P}, C_1 || C_2 || \overline{C})$
 $D \leftarrow R \leftarrow \emptyset$, where D, R are multisets
for $r = 1$ to q **do**
 if $qt^r = \text{Enc}$ **then**
 $X_1^r \xleftarrow{\$} \{0, 1\}^{2n}$
 $A_1^r \leftarrow \text{MSB}_n(X_1^r)$, $A_2^r \leftarrow \text{LSB}_n(X_1^r)$
 $F^r \xleftarrow{\$} \{0, 1\}^n$; $B^r \xleftarrow{\$} \{0, 1\}^n$; $J^r = F^r \oplus B^r$
 $D \leftarrow D \cup \{P_1^r \oplus A_2^r\}$; $D \leftarrow D \cup \{P_2^r \oplus A_1^r \oplus F^r\}$
 for $i = 0$ to $\lfloor |\overline{P}^r|/n \rfloor$ **do**
 $D \leftarrow D \cup \{J^r \oplus i\}$
 end for
 $X_2^r \xleftarrow{\$} \{0, 1\}^{2n}$
 $A_3^r \leftarrow \text{MSB}_n(X_2^r)$, $A_4^r \leftarrow \text{LSB}_n(X_2^r)$
 $D \leftarrow D \cup \{B^r \oplus A_3^r\}$; $D \leftarrow D \cup \{F^r \oplus A_4^r \oplus C_2^r\}$
 else if $qt^r = \text{Dec}$ **then**
 $\overline{X}_1^r \xleftarrow{\$} \{0, 1\}^{2n}$
 $\overline{A}_1^r \leftarrow \text{MSB}_n(\overline{X}_1^r)$, $\overline{A}_2^r \leftarrow \text{LSB}_n(\overline{X}_1^r)$
 $\overline{F}^r \xleftarrow{\$} \{0, 1\}^n$; $\overline{B}^r \xleftarrow{\$} \{0, 1\}^n$; $\overline{J}^r \leftarrow \overline{F}^r \oplus \overline{B}^r$
 $D \leftarrow D \cup \{\overline{F}^r \oplus \overline{A}_2^r \oplus C_2^r\}$; $D \leftarrow D \cup \{\overline{B}^r \oplus \overline{A}_1^r\}$
 for $i = 0$ to $\lfloor |\overline{C}^r|/n \rfloor$ **do**
 $D \leftarrow D \cup \{\overline{J}^r \oplus i\}$
 end for
 $\overline{X}_2^r \xleftarrow{\$} \{0, 1\}^{2n}$
 $\overline{A}_3^r \leftarrow \text{MSB}_n(\overline{X}_2^r)$, $\overline{A}_4^r \leftarrow \text{LSB}_n(\overline{X}_2^r)$
 $D \leftarrow D \cup \{P_1^r \oplus \overline{A}_4^r\}$; $D \leftarrow D \cup \{P_2^r \oplus \overline{A}_3^r \oplus \overline{F}^r\}$
 end if
end for
 $\text{bad} \leftarrow \text{true}$ **if** (some value appears more than once in D)

The probability of the *bad* event occurring in **Game6**, denoted as $\Pr[\text{Game6 sets } \textit{bad}]$, is bounded by the probability of a collision occurring within the set D , as evident in the (implicit) pseudocode of **Game6**. Therefore, to compute this collision probability, it is necessary to consider all elements that are added to the set D throughout the game's execution.

$$\begin{aligned} D = & \{(P_2 \oplus A_1 \oplus F)^r \mid qt^r = \textit{Enc}\} \cup \{(P_1 \oplus A_2)^r \mid qt^r = \textit{Enc}\} \\ & \cup \{(J \oplus i)^r, \forall i \in [0, \lfloor \overline{P}^r \rfloor / n] \mid qt^r = \textit{Enc}\} \cup \{(B \oplus A_3)^r \mid qt^r = \textit{Enc}\} \\ & \cup \{(F \oplus A_4 \oplus C_2)^r \mid qt^r = \textit{Enc}\} \cup \{(\overline{B} \oplus \overline{A_1})^r \mid qt^r = \textit{Dec}\} \\ & \cup \{(\overline{F} \oplus \overline{A_2} \oplus C_2)^r \mid qt^r = \textit{Dec}\} \cup \{(\overline{J} \oplus i)^r, \forall i \in [0, \lfloor \overline{C}^r \rfloor / n] \mid qt^r = \textit{Dec}\} \\ & \cup \{(P_2 \oplus \overline{A_3} \oplus \overline{F})^r \mid qt^r = \textit{Dec}\} \cup \{(P_1 \oplus \overline{A_4})^r \mid qt^r = \textit{Dec}\} \end{aligned}$$

The following enumerate all potential collision scenarios within the set D :

- The probability of collisions between terms $(P_2 \oplus \overline{A_3} \oplus \overline{F})$, $(P_1 \oplus \overline{A_4})$ originating from decryption queries and $(P_2 \oplus A_1 \oplus F)$, $(P_1 \oplus A_2)$ arising from encryption queries is dependent upon the randomness of $\overline{A_3}$, $\overline{A_4}$, $\overline{F}$, A_1 , A_2 and F . As these terms are independent random values introduced within the game's model, they are the sole source of potential collision for these terms. As a result, the collision probability for these specific cases is 2^{-n} .
- The terms $(B \oplus A_3)$, $(F \oplus A_4 \oplus C_2)$ from encryption queries and $(\overline{B} \oplus \overline{A_1})$, $(\overline{F} \oplus \overline{A_2} \oplus C_2)$ from decryption queries are derived from random choices J , F , A_3 , A_4 , $\overline{J}$, $\overline{A_1}$, $\overline{A_2}$, and $\overline{F}$, which are independent of the adversary's queries and the corresponding responses. Consequently, the probability of a collision in these cases is 2^{-n} , as it solely depends on the likelihood of these independent random values coinciding.
- Similarly, the terms $(J \oplus i)$ in encryption queries and $(\overline{J} \oplus i)$ in decryption queries are also based on random choices. Therefore, the collision probability in these instances is again 2^{-n} .

Consequently, the overall collision probability within the set D is bounded by $\binom{q(a+2)}{2} \cdot 2^{-n}$, since there are at most $q(a+2)$ variables added to D , each having a collision probability of 2^{-n} . As a result, we conclude that $\Pr[\text{Game6 sets } \textit{bad}] \leq \binom{q(a+2)}{2} \cdot 2^{-n}$. Combining this inequality with the

previous results, we arrive at the final bound:

$$\begin{aligned}
\mathbf{Adv}_{\text{Animagus}[E,H]}^{SPRP} &\leq \mathbf{Adv}_{\text{Animagus}[P,F]}^{SPRP} + 2 \cdot \mathbf{Adv}_E^{SPRP} + \frac{\binom{q}{2}}{2^{2n}} \\
&\leq \mathbf{Adv}_{\text{Animagus}[P,F]}^{SPRF} + 2 \cdot \mathbf{Adv}_E^{SPRP} + \frac{q^2}{2^{n+1}} + \frac{\binom{q}{2}}{2^{2n}} \\
&\leq 2 \cdot \Pr[\text{Game6 sets bad}] + \frac{\binom{q}{2}}{2^n} + 2 \cdot \mathbf{Adv}_E^{SPRP} + \frac{q^2}{2^{n+1}} + \frac{\binom{q}{2}}{2^{2n}} \\
&\leq 2 \cdot \mathbf{Adv}_E^{SPRP} + 2 \cdot \frac{\binom{q(a+2)}{2}}{2^n} + \frac{\binom{q}{2}}{2^n} + \frac{q^2}{2^{n+1}} + \frac{\binom{q}{2}}{2^{2n}} \\
&\leq 2 \cdot \mathbf{Adv}_E^{SPRP} + \frac{\binom{q(a+2)}{2}}{2^{n-1}} + \frac{\binom{q}{2}}{2^n} + \frac{q^2}{2^{n+1}} + \frac{\binom{q}{2}}{2^{2n}}
\end{aligned}$$

□

4.1 Beyond-Birthday-Bound Security

Beyond-birthday-bound security refers to the ability of a cryptographic mode of operation to maintain its security properties against generic attacks, particularly collision-finding attacks, even when the total number of processed blocks of data significantly exceeds the conventional birthday paradox limit. Achieving security beyond this point is critical for applications that handle vast amounts of data, such as real-time video streaming, large-scale database encryption, and secure cloud storage. While highly desirable for specific use cases, implementing beyond-birthday-bound security often introduces significant performance and coding overhead. Given these trade-offs, our design philosophy prioritizes a general-purpose solution with robust birthday-bound security. This approach offers an optimal balance between security and performance for a broad range of applications where the data volume does not necessitate BBB security. Acknowledging the importance of supporting data-intensive applications, we plan to release an updated version of our design that incorporates the necessary mechanisms to provide beyond-birthday-bound security for those specific use cases.

To achieve BBB security, the proposed design will be improved with architectural refinements derived from established high-throughput frameworks, such as Iwata’s CENC [21] and Mennink’s rekeying paradigms [23]. The updated BBB secure version will aim to reduce collision vulnerabilities by deriving subkeys to ensure that the cryptographic components operate under independent keys. Furthermore, for the counter mode implementation, we enhance the security margin by suppressing the internal collision

probability. This is achieved through a computationally efficient masking technique, strategically designed to push the adversarial advantage beyond the standard $O(2^{n/2})$ birthday limit.

4.2 Supplementary Security

In addition to the fundamental security review of the algorithm, this section will cover security properties for applications based on derived functions. The security framework integrates NIST-recommended methodologies to ensure robust protection; accordingly, these features and their corresponding standards are defined herein.

4.2.1 Nonce Misuse Resistant

By incorporating a nonce as a derived tweak via a keyed hash function, cryptographic systems can achieve nonce misuse resistance. This architecture ensures that the encryption process remains secure even if nonces are reused, provided the associated data or keying material varies. However, the mechanism remains deterministic: identical inputs (nonce, key, and associated data) will inevitably yield identical tweaks and ciphertexts.

Consequently, this process provides a robust level of misuse resistance for the proposed design. The variability introduced by the tweak, which is a function of the nonce and other inputs, prevents predictable outcomes even when identical nonce values are supplied. This design mitigates the catastrophic failures typically associated with nonce reuse, bolstering the overall integrity and confidentiality of the cryptographic primitive.

4.2.2 Nonce Hiding

In specific cryptographic applications, when a nonce is derived from private data, it becomes a sensitive value that requires concealment. This security property is termed nonce hiding. Nonce hiding is a security feature provided in our design by processing the nonce with a keyed hash function. This mechanism effectively prevents unauthorized discovery of the nonce without access to the corresponding private key. In scenarios where sensitive information, such as a counter or a timestamp, is used as a nonce, this technique ensures its confidentiality, thereby mitigating potential information leakage or traffic analysis attacks.

4.2.3 Release of Unverified Plaintext

Incorrect use of cryptographic primitives can introduce significant vulnerabilities, such as the Release of Unverified Plaintext (RUP). This vulnerability can manifest even in the absence of a formal authentication mechanism. The unverified data, although seemingly random, may be structured and exploitable.

To mitigate this, standards bodies such as NIST recommend the encode-then-encrypt method. By authenticating messages prior to their release, this approach achieves the security property of RUP resistance. This design ensures that any data released to a user is either legitimate and correctly authenticated or entirely random and cryptographically unusable, thereby preventing an attacker from gathering useful information from the unverified output.

5 Conclusion

Our Animagus design offers a versatile and secure approach to variable-length block cipher operation. We’ve tackled several key limitations of traditional block cipher modes, particularly the rigid requirement for padding. By accommodating variable-length inputs without padding, Animagus provides a more efficient and user-friendly solution for a wide range of applications. This is a significant advantage over many existing schemes that necessitate padding, which can introduce overhead and complexity.

A key architectural choice in our design is the hash-encrypt-hash structure, inspired by the NIST-recommended HCTR2 model. This structure balances performance with flexibility, making it a robust choice for a general-purpose mode of operation. A notable departure from schemes using a universal hash is our use of a cryptographic hash function, specifically HMAC. While this introduces a minor performance trade-off, it significantly enhances security by providing collision resistance. This decision aligns with the goal of creating a highly secure and adaptable framework. We also included a secondary encryption branch to facilitate authenticated encryption, a feature that further extends Animagus’s utility without compromising its core security principles.

We designed Animagus with birthday-bound security to ensure it remains a practical and flexible solution. While beyond-birthday-bound security is valuable for certain applications, the associated performance overhead and implementation complexity make it less suitable for many common use cases, especially with larger block sizes. Our choice prioritizes a balance of strong

security and broad applicability, making Animagus a versatile candidate for diverse cryptographic needs. We recognize that future work could explore a beyond-birthday-bound version of Animagus for specialized high-security environments.

In this paper, we proposed and formally analyzed Animagus, a novel block cipher mode of operation that addresses the limitations of fixed-length block cipher modes. Our design offers a flexible, padding-free solution for variable-length inputs while maintaining robust pseudorandom permutation security. The hash-encrypt-hash structure, combined with the use of a cryptographic hash function, ensures a high level of security and adaptability with a wide range of underlying block ciphers. Animagus’s ability to handle variable-length inputs and tweaks, as well as its provision for authenticated encryption, makes it a more versatile and efficient alternative to many existing schemes. Through our formal security proof, we have established that Animagus achieves a high level of security without sacrificing practicality, making it a strong candidate for future cryptographic applications that demand a blend of flexibility and strong security guarantees.

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